Lab Assignment 1
U24AI022

Q1 Consider a map given in image.

Implement Breadth First Search(BFS) and Depth First Search (DFS) algorithm to find all possible step cost between Syracuse to Chicago.

CODE:

```
#include<bits/stdc++.h>
using namespace std;
struct p{
    string s;
    vector<string> path;
    int cost;
    unordered_set<string> visited;
};
void bfs(string src, string
dest,unordered_map<string,vector<pair<string,int>>> &g){
    queue<p> q;
    q.push({src,{src},0,{src}});
    while(!q.empty()){
        p curr=q.front();
        q.pop();
        if(curr.s==dest){
            for(auto city : curr.path){
                cout<<city<<"--->";
            }
```

```
            cout<<"total_path_cost:"<<curr.cost<<endl;
            continue;
        }
        for(auto neigh : g[curr.s]){
            string next_city=neigh.first;
            int c=neigh.second;
            if(!curr.visited.count(next_city)){
                p next_q_ele;
                next_q_ele.s=next_city;
                next_q_ele.path=curr.path;
                next_q_ele.path.push_back(next_city);
                next_q_ele.cost=curr.cost;
                next_q_ele.cost+=c;
                next_q_ele.visited=curr.visited;
                next_q_ele.visited.insert(next_city);
                q.push(next_q_ele);
            }
        }
    }
}

int bfs_exp_path(string src,string
dest,unordered_map<string,vector<pair<string,int>>> &g){
    queue<p> q;
    q.push({src,{src},0});
    unordered_set<string> visited;
    visited.insert(src);
    int total_exp_cost=0;
    while(!q.empty()){
        p curr=q.front();
        q.pop();
        if(curr.s==dest){
            break ;
        }
        total_exp_cost+=curr.cost;
        for(auto neigh : g[curr.s]){
            string next_city=neigh.first;
            int c=neigh.second;
            if(!visited.count(next_city)){
                p next_q_ele;
                visited.insert(next_city);
                next_q_ele.s=next_city;
                next_q_ele.path=curr.path;
                next_q_ele.path.push_back(next_city);
                next_q_ele.cost=curr.cost+c;
```

```
                q.push(next_q_ele);
            }
        }
    }
    return total_exp_cost;
}

void dfs(string curr, string
dest,unordered_map<string,vector<pair<string,int>>> &g
,unordered_set<string> &visited ,vector<string> &dfs_path ,int cost){
    if(curr==dest){
        for(auto city : dfs_path){
            cout<<city<<"--->";
        }
        cout<<"total_path_cost:"<<cost<<endl;
        return;
    }
    for(auto &neigh : g[curr]){
        string next_city=neigh.first;
        int c=neigh.second;
        if(!visited.count(next_city)){
            visited.insert(next_city);
            dfs_path.push_back(next_city);
            dfs(next_city,dest,g,visited,dfs_path,cost+c);
            dfs_path.pop_back();
            visited.erase(next_city);
        }
    }
}

int main(){
    unordered_map<string,vector<pair<string,int>>> g;
    g["chicago"] =
{{"detroit",283},{"cleveland",345},{"indianapolis",182}};
    g["indianapolis"] = {{"chicago",182},{"columbus",176}};
    g["cleveland"] =
{{"chicago",345},{"detroit",169},{"columbus",144},{"pittsburgh",134},{"b
uffalo",189}};
    g["detroit"] = {{"chicago",283},{"cleveland",169},{"buffalo",256}};
    g["columbus"] =
{{"indianapolis",176},{"pittsburgh",185},{"cleveland",144}};
    g["pittsburgh"] =
{{"cleveland",134},{"columbus",185},{"buffalo",215},{"philadelphia",305}
,{"baltimore",247}};
    g["baltimore"] = {{"pittsburgh",247},{"philadelphia",101}};
```

```
    g["philadelphia"] = {{"baltimore",101},{"pittsburgh",305},{"new
york",97},{"syracuse",253}};
    g["buffalo"] =
{{"detroit",256},{"cleveland",189},{"pittsburgh",215},{"syracuse",150}};
    g["new york"] =
{{"philadelphia",97},{"syracuse",254},{"boston",215},{"providence",181}}
;
    g["providence"] = {{"new york",181},{"boston",50}};
    g["boston"] = {{"syracuse",312},{"providence",50},{"new
york",215},{"portland",107}};
    g["portland"] = {{"boston",107}};
    g["syracuse"] = {{"buffalo",150},{"philadelphia",253},{"new
york",254},{"boston",312}};

    string src="syracuse" , dest="chicago";
    cout<<"BFS TO FIND ALL POSSIBLE PATHS BETWEEN SRC AND DEST"<<endl;
    bfs(src,dest,g);
    cout<<"DFS TO FIND ALL POSSIBLE PATHS BETWEEN SRC AND DEST"<<endl;\
    unordered_set<string> visited;
    vector<string> dfs_path;
    visited.insert(src);
    dfs_path.push_back(src);
    dfs(src,dest,g,visited,dfs_path,0);
}
```

OUTPUT:
BFS TO FIND ALL POSSIBLE PATHS BETWEEN SRC AND DEST
syracuse--->buffalo--->detroit--->chicago--->total_path_cost:689
syracuse--->buffalo--->cleveland--->chicago--->total_path_cost:684
syracuse--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:920
syracuse--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:791
syracuse--->buffalo--->pittsburgh--->cleveland--->chicago--->total_path_cost:844
syracuse--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1037
syracuse--->buffalo--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:841
syracuse--->buffalo--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:951
syracuse--->buffalo--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:908
syracuse--->buffalo--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1039
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_cost:1080
syracuse--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1144
syracuse--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1101
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_pa

th_cost:1232
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1312
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1307
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1135
syracuse--->buffalo--->detroit--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1077
syracuse--->buffalo--->cleveland--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1016
syracuse--->buffalo--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1001
syracuse--->buffalo--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1146
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1187
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1144
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1275
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1355
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1350
syracuse--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1194
syracuse--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1420
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1339
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1543
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1414
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_cost:1178
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1242
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1199
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1330
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1410
syracuse--->new

york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1405
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1408
syracuse--->buffalo--->detroit--->cleveland--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1252
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1237
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1463
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1382
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1586
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1457
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1615
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1464
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1285
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1242
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1373
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1453
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1448
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1292
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1518
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1437
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1641
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1512
syracuse--->boston--->providence--->new

york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1424
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_c
ost:1451
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost
:1515
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path
_cost:1472
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_c
ost:1603
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:16
83
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost
:1678
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffal
o--->detroit--->chicago--->total_path_cost:1658
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbu
s--->indianapolis--->chicago--->total_path_cost:1507
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus---
>indianapolis--->chicago--->total_path_cost:1700
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapoli
s--->chicago--->total_path_cost:1335
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chic
ago--->total_path_cost:1561
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->c
hicago--->total_path_cost:1480
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chic
ago--->total_path_cost:1684
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chic
ago--->total_path_cost:1555
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:1713
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1562
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_c
ost:1467
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost

:1531
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path
_cost:1488
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_c
ost:1619
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:16
99
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost
:1694
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->tot
al_path_cost:1558
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago
--->total_path_cost:1515
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->
total_path_cost:1646
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_
path_cost:1726
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->tot
al_path_cost:1721
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago
--->total_path_cost:1565
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_
path_cost:1791
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->tot
al_path_cost:1710
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_
path_cost:1914
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_
path_cost:1785
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland---
>columbus--->indianapolis--->chicago--->total_path_cost:1743
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->
detroit--->chicago--->total_path_cost:1756
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->i
ndianapolis--->chicago--->total_path_cost:1605

syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:1798
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->tot
al_path_cost:1574
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago
--->total_path_cost:1531
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->
total_path_cost:1662
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_
path_cost:1742
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->tot
al_path_cost:1737
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago
--->total_path_cost:1581
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_
path_cost:1807
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->tot
al_path_cost:1726
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_
path_cost:1930
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_
path_cost:1801
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapoli
s--->chicago--->total_path_cost:1608
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chic
ago--->total_path_cost:1834
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->c
hicago--->total_path_cost:1753
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chic
ago--->total_path_cost:1957
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chic
ago--->total_path_cost:1828
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:1986

syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1835
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->colu
mbus--->indianapolis--->chicago--->total_path_cost:1841
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapoli
s--->chicago--->total_path_cost:1624
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chic
ago--->total_path_cost:1850
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->c
hicago--->total_path_cost:1769
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chic
ago--->total_path_cost:1973
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chic
ago--->total_path_cost:1844
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:2002
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1851
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->
detroit--->chicago--->total_path_cost:2029
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->i
ndianapolis--->chicago--->total_path_cost:1878
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:2071
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->
detroit--->chicago--->total_path_cost:2045
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->i
ndianapolis--->chicago--->total_path_cost:1894
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:2087
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->colu
mbus--->indianapolis--->chicago--->total_path_cost:2114
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->colu
mbus--->indianapolis--->chicago--->total_path_cost:2130

DFS TO FIND ALL POSSIBLE PATHS BETWEEN SRC AND DEST

syracuse--->buffalo--->detroit--->chicago--->total_path_cost:689
syracuse--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:920
syracuse--->buffalo--->detroit--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1077
syracuse--->buffalo--->detroit--->cleveland--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1252
syracuse--->buffalo--->cleveland--->chicago--->total_path_cost:684
syracuse--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:791
syracuse--->buffalo--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:841
syracuse--->buffalo--->cleveland--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1016
syracuse--->buffalo--->pittsburgh--->cleveland--->chicago--->total_path_cost:844
syracuse--->buffalo--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:951
syracuse--->buffalo--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1001
syracuse--->buffalo--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:908
syracuse--->buffalo--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1039
syracuse--->buffalo--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1146
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_cost:1080
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1187
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1237
syracuse--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1463
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1144
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1275
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1382
syracuse--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1658
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1355
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1586
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1743
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1350
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1457
syracuse--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbu

s--->indianapolis--->chicago--->total_path_cost:1507
syracuse--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1037
syracuse--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1144
syracuse--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1194
syracuse--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1420
syracuse--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1101
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1232
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1339
syracuse--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1615
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1312
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1543
syracuse--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1700
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1307
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1414
syracuse--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1464
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_cost:1178
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1285
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1335
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1561
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1242
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1373
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1480
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->

detroit--->chicago--->total_path_cost:1756
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_
path_cost:1453
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chic
ago--->total_path_cost:1684
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->colu
mbus--->indianapolis--->chicago--->total_path_cost:1841
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->tot
al_path_cost:1448
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chic
ago--->total_path_cost:1555
syracuse--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->i
ndianapolis--->chicago--->total_path_cost:1605
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1135
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost
:1242
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago
--->total_path_cost:1292
syracuse--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_
path_cost:1518
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path
_cost:1199
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_c
ost:1330
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->tot
al_path_cost:1437
syracuse--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:1713
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:14
10
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_
path_cost:1641
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:1798

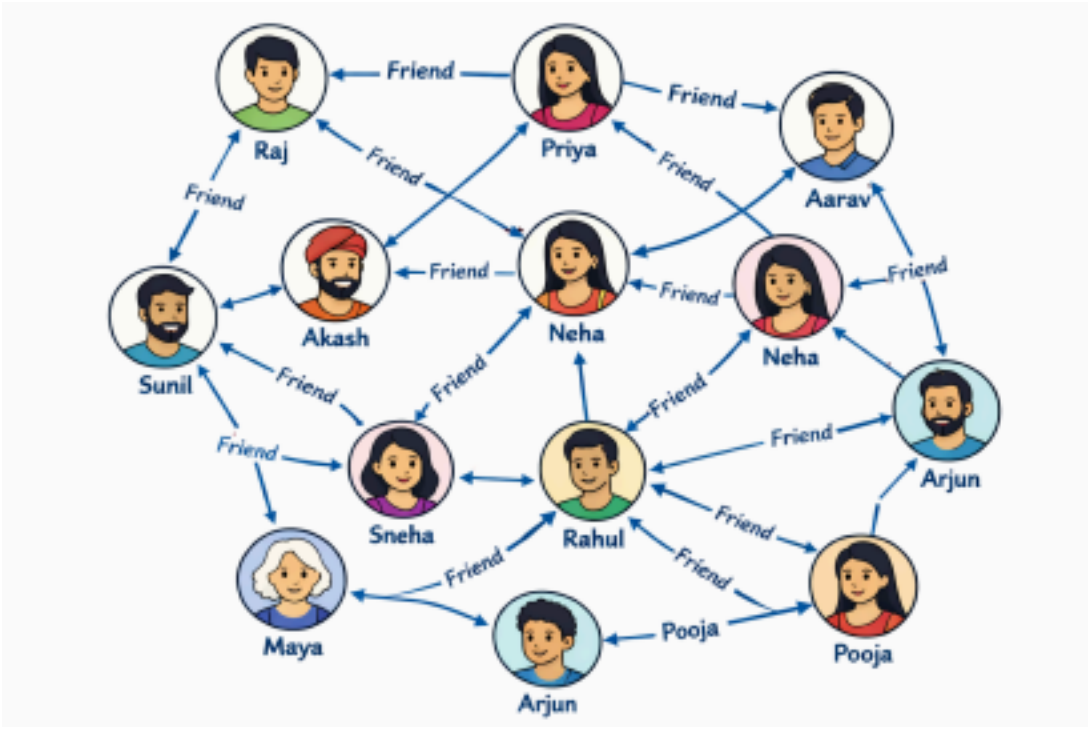
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost
:1405
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_
path_cost:1512
syracuse--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1562
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_c
ost:1467
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->tot
al_path_cost:1574
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapoli
s--->chicago--->total_path_cost:1624
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chic
ago--->total_path_cost:1850
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago
--->total_path_cost:1531
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->
total_path_cost:1662
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->c
hicago--->total_path_cost:1769
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->
detroit--->chicago--->total_path_cost:2045
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_
path_cost:1742
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chic
ago--->total_path_cost:1973
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->colu
mbus--->indianapolis--->chicago--->total_path_cost:2130
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->tot
al_path_cost:1737
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chic
ago--->total_path_cost:1844
syracuse--->boston--->providence--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->i
ndianapolis--->chicago--->total_path_cost:1894

syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1424
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost
:1531
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago
--->total_path_cost:1581
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_
path_cost:1807
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path
_cost:1488
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_c
ost:1619
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->tot
al_path_cost:1726
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:2002
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:16
99
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_
path_cost:1930
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:2087
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost
:1694
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_
path_cost:1801
syracuse--->boston--->providence--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1851
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->chicago--->total_path_c
ost:1451
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->detroit--->chicago--->tot
al_path_cost:1558
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->columbus--->indianapoli
s--->chicago--->total_path_cost:1608
syracuse--->boston--->new

york--->philadelphia--->baltimore--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1834
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1515
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1646
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->total_path_cost:1753
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:2029
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:1726
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_path_cost:1957
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:2114
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost:1721
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_path_cost:1828
syracuse--->boston--->new
york--->philadelphia--->baltimore--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1878
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->chicago--->total_path_cost:1408
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->detroit--->chicago--->total_path_cost:1515
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->columbus--->indianapolis--->chicago--->total_path_cost:1565
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->cleveland--->buffalo--->detroit--->chicago--->total_path_cost:1791
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->indianapolis--->chicago--->total_path_cost:1472
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->chicago--->total_path_cost:1603
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->detroit--->chicago--->tot

al_path_cost:1710
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->columbus--->cleveland--->buffalo--->detroit--->chi
cago--->total_path_cost:1986
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->chicago--->total_path_cost:16
83
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->chicago--->total_
path_cost:1914
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->detroit--->cleveland--->columbus--->indi
anapolis--->chicago--->total_path_cost:2071
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->chicago--->total_path_cost
:1678
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->detroit--->chicago--->total_
path_cost:1785
syracuse--->boston--->new
york--->philadelphia--->pittsburgh--->buffalo--->cleveland--->columbus--->indianapolis--
->chicago--->total_path_cost:1835
BFS TOTAL EXPLORATION PATH BETWEEN SRC AND DEST
3287

Q2 Consider a social media network as given below. Find BFS and DFS tree.



CODE:

```
#include<bits/stdc++.h>
```

```
using namespace std;
vector<string> bfs(unordered_map<string,vector<string>> &g){
    int n=g.size();
    vector<string> ans;
    unordered_map<string,bool> visited;
    queue<string> q;
    ans.reserve(n);
    string src="raj";
    visited[src]=true;
    q.push(src);
    while(!q.empty()){
        string s=q.front();
        q.pop();
        ans.emplace_back(s);
        for(auto v : g[s]){
            if(!visited[v]){
                q.push(v);
                visited[v]=true;
            }
        }
    }
    return ans;
}

void dfs1(unordered_map<string,vector<string>>
&g,unordered_map<string,bool> &visited,string &s,vector<string> &ans){
    visited[s]=true;
    ans.emplace_back(s);
    for(auto cv : g[s]){
        if(visited[cv]==false){
            dfs1(g,visited,cv,ans);
        }
    }
}
vector<string> dfs(unordered_map<string,vector<string>> &g){
    int n=g.size();
    vector<string> ans;
    unordered_map<string,bool> visited;
    ans.reserve(n);
    for(auto v : g){
        string s=v.first;
        if(visited[s]==false){
            dfs1(g,visited,s,ans);
        }
    }
```

```
    return ans;
}

int main(){
    unordered_map<string,vector<string>> g;
    g["raj"]={"neha","sunil"};
    g["akash"]={"sunil","priya"};
    g["sunil"]={"raj","maya","sneha","akash"};
    g["sneha"]={"sunil","rahul","neha",};
    g["maya"]={"sunil","rahul","arjun1"};
    g["priya"]={"raj","neha","aarav","akash"};
    g["neha"]={"raj","akash","aarav","sneha"};
    g["neha1"]={"priya","neha","rahul","aarav"};
    g["rahul"]={"neha","neha1","sneha","pooja","maya","arjun"};
    g["aarav"]={"neha","neha1","arjun"};
    g["arjun"]={"rahul","neha1","aarav"};
    g["arjun1"]={"maya","pooja"};
    g["pooja"]={"rahul","arjun","arjun1"};

    vector<string> result1=bfs(g);
    cout<<"BFS"<<endl;
    for(auto i : result1) cout<<i<<" ";
    cout<<endl;
    vector<string> result2=dfs(g);
    cout<<"DFS"<<endl;
    for(auto i  : result2) cout<<i<<" ";
}
```

OUTPUT:

BFS:

raj neha sunil akash aarav sneha maya priya neha1 arjun rahul arjun1 pooja

DFS:

arjun1 maya sunil raj neha akash priya aarav neha1 rahul sneha pooja arjun